

Multi-Wavelength Analysis of Solar Limb Darkening and Photospheric Temperature Structure

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Abstract

Solar limb darkening describes the decrease in observed intensity from the centre of the solar disc towards the limb and provides a direct probe of the thermal structure of the solar photosphere. In this project, drift scan observations of the Sun were performed at multiple wavelengths using a refracting telescope and CCD imaging system. Centre to limb intensity profiles were extracted from the recorded data and modelled using parametrised limb darkening functions fitted via least squares methods.

The fitted limb darkening coefficients were used, together with the Eddington–Barbier approximation and an assumption of local thermal equilibrium, to reconstruct the source function and derive absolute temperature profiles as a function of optical depth. The resulting temperature distributions show a systematic decrease towards smaller optical depths and exhibit clear wavelength dependence, consistent with standard models of radiative transfer in stellar atmospheres. Despite experimental limitations arising from atmospheric seeing and detector noise, the results demonstrate that drift scan observations provide a simple and effective method for investigating the thermal structure of the solar photosphere.

1 Introduction

The intensity of radiation emerging from the solar surface of our Sun is not uniform across the visible disc, but rather decreases from the centre towards the limb; an effect known as solar limb darkening. This phenomenon arises because radiation observed at different viewing angles originates from different depths within the solar photosphere, where the temperature varies with optical depth as the number of photons escaping varies due to multiple factors. Measurements of limb darkening therefore provide a direct observational probe of the thermal structure of the solar atmosphere.

In this project, observations of the Sun were carried out at multiple wavelengths using a refracting telescope and CCD imaging system. The centre to limb intensity variation was measured using a drift scan method, allowing the solar intensity profile to be sampled while avoiding instrumental vignetting effects. The measured intensity profiles were modelled using a parameter focused limb darkening function, with best-fit parameters determined via the method of least squares.

From the fitted intensity profiles, the source function $S(\tau)$ was reconstructed as a function of optical depth for each wavelength. Assuming local thermal equilibrium, the source function was then used to infer the temperature structure of the solar photosphere. The resulting temperature profiles were analysed and discussed in the context of standard solar atmosphere models, with particular attention paid to wavelength dependence and experimental

uncertainties.

2 Observations and Data

2.1 Telescope and Instrumentation

Observations of the Sun were carried out using a Skywatcher Equinox 80 mm refracting telescope with a focal length of 500 mm, corresponding to an f -ratio of 6.25. The refractor was mounted on the PlaneWave CDK 400 equatorial mount, providing stable alignment with the Earth’s rotation axis and allowing controlled tracking behaviour during observations. A certified Baader AstroSolar neutral density solar filter, with a transmission of approximately 10^{-5} , was fitted to the front of the refracting telescope to enable safe solar observation and to protect the detector from excessive radiation.

2.2 CCD Camera and Filters

Images were recorded using a ZWO ASI462MM monochrome CMOS camera, designed for high-resolution planetary and solar imaging. The camera has a resolution of 1936×1096 pixels, a pixel size of $2.9 \mu\text{m}$, and a 12-bit dynamic range. Data acquisition was carried out using the SharpCap software installed on the observatory computer. Exposure time and gain settings were manually adjusted to ensure a high signal level while avoiding pixel saturation, and automatic control settings were disabled to maintain consistency throughout the observations.

Spectral filtering was achieved using a ZWO EFW Mini electronic filter wheel, allowing rapid selection between different wavelength bands. Baader UV, Blue, Green, and Red filters were used to measure the limb darkening effect at multiple wavelengths. These filters have representative central wavelengths of approximately 350 nm, 445 nm, 538 nm, and 638 nm respectively. Observing at multiple wavelengths enables investigation of the wavelength dependence of limb darkening and provides information on the thermal structure of the solar photosphere.

2.3 Observation Method and Data Acquisition

During observations, the apparent motion of the Sun across the field of view caused by the Earth’s rotation was exploited using a drift scan method. Telescope tracking was switched off, allowing the solar disc to drift naturally across the CCD sensor. This method avoids the need to correct for vignetting effects across the full field of view, as intensity measurements were taken from a fixed central region of the detector while the Sun passed through it.

Images were captured at a cadence of one frame per second, producing approximately 200 images per filter during each drift scan. All images were saved in FITS format to preserve the full

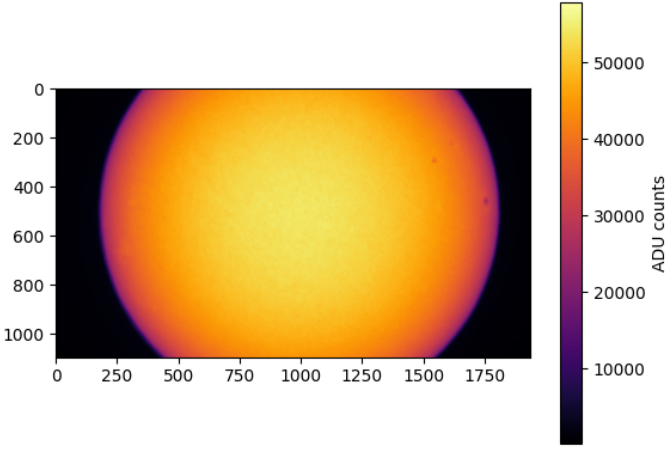


Figure 1: Image of the Sun taken with the aforementioned equipment; observed utilising the green filter.

dynamic range and associated metadata. The data were later transferred for processing and analysis using Python, where pixel intensities were extracted and associated with corresponding positions on the solar disc.

Observations were conducted under clear sky conditions with minimal cloud cover and no visible obstructions to the solar disc. Atmospheric conditions remained stable throughout the observing period, allowing clean imaging of the Sun with no significant artefacts affecting the recorded intensity profiles.

3 Data Processing and Uncertainties

The recorded FITS images were processed using Python with the `Astropy` and `NumPy` libraries. For each drift scan, intensity measurements were extracted from a single fixed pixel located near the centre of the CCD array. This pixel was chosen to ensure a consistent detector response throughout the scan and to avoid edge effects or vignetting present elsewhere in the field of view.

As the telescope tracking was switched off, the apparent motion of the Sun across the detector caused the solar disc to drift across this fixed pixel. In the initial and final frames of each scan, the pixel recorded only background signal, while during the central portion of the scan the pixel recorded solar intensity as the disc passed across it. This behaviour produced a characteristic intensity profile with low counts outside the solar disc and a pronounced peak when the Sun intersected the selected pixel.

The resulting drift scan intensity profile is shown in Figure 2, illustrating the background level outside the solar disc and the sharp intensity gradients at the solar limb.

The position on the solar disc was parametrised by a dimensionless coordinate,

$$x = \frac{r}{R} \quad (1)$$

where r is the projected distance from the disc centre and R is the solar radius. Rather than determining r geometrically from the image, the coordinate x was derived from the frame index of the drift scan. The frames corresponding to the solar limb were identified empirically as those exhibiting the steepest intensity gradients, corresponding to the moments when the Sun

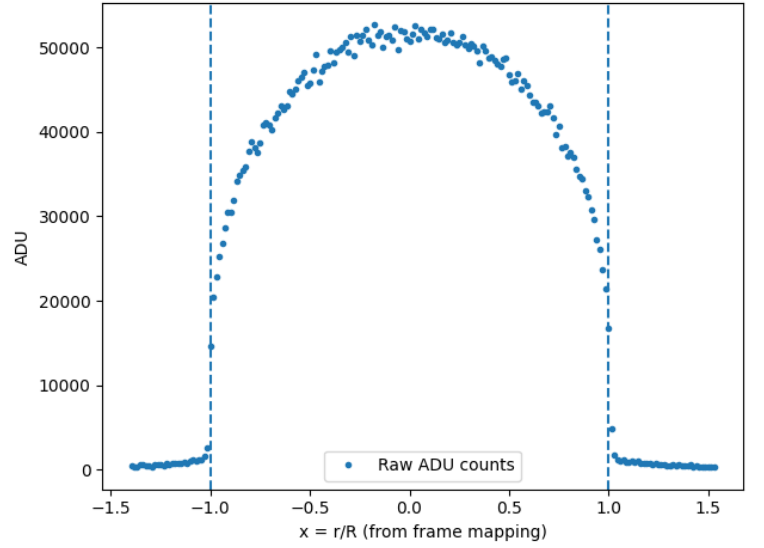


Figure 2: Raw intensity values recorded at a fixed central pixel as a function of the mapped disc coordinate $x = r/R$ during a drift scan. Low count regions correspond to frames where the solar disc does not intersect the selected pixel, while the central region shows the passage of the solar disc. The dashed lines indicate the empirically identified limb positions used to define the interval $-1 \leq x \leq 1$.

entered and exited the selected pixel. These were found to occur at frames 27 and 163 respectively. Using these limits, the frame index was linearly mapped onto the interval $-1 \leq x \leq 1$, with $x = 0$ corresponding to the centre of the solar disc.

This procedure resulted in a set of measured intensities $I(x)$ sampled across the solar disc for each wavelength. The same processing method was applied consistently to all filters.

Uncertainties in the measured intensities were estimated by examining local variations in pixel values surrounding the selected central pixel. For each frame, the standard deviation of pixel values within a narrow vertical strip centred on the chosen pixel was computed and taken as the uncertainty σ_i associated with that intensity measurement. This approach captures contributions from photon noise and detector noise while remaining insensitive to large scale gradients across the image.

Background signal arising from sky brightness, scattered light, and instrumental offsets was not explicitly subtracted at this stage, but was instead accounted for during the fitting procedure through the inclusion of a constant background term in the model. The resulting intensity measurements and associated uncertainties were then used in the least squares fitting of the limb darkening profiles.

4 Limb Darkening Fits

For each filter, the centre to limb intensity variation was obtained as a set of measured intensity values $I(x)$ sampled across the solar disc during the drift scan. To remove the dependence on absolute detector response, the intensity profile was normalised by the maximum recorded intensity during the scan, taken as a proxy for the disc centre intensity $I(0)$. This produced a dimensionless profile $I(x)/I(0)$ suitable for comparison with analytical limb darkening models.

The measured intensity profiles were fitted using the parametrised limb darkening function of the form

$$\frac{I(x)}{I(0)} = a_0 + a_1 \sqrt{1 - x^2} + (1 - a_0 - a_1)(1 - x^2), \quad (2)$$

where $x = r/R$ is the normalised disc coordinate, r is the projected distance from the centre of the solar disc, and R is the solar radius. The parameters a_0 and a_1 control the strength and shape of the limb darkening effect, while the quadratic coefficient is constrained by the normalisation condition at the disc centre.

Best fit values of the parameters were obtained using a non-linear least-squares fitting procedure implemented with the `scipy.optimize.curve_fit` routine. The uncertainties on the fitted parameters were estimated from the square roots of the diagonal elements of the covariance matrix returned by the fitting algorithm.

Figure 3 shows the normalised intensity profile for the Blue filter together with the best-fit limb darkening model. The fitted curve reproduces the overall trend of the data well, with a clear decrease in intensity towards the solar limb. Small deviations between the data and the fitted model are attributed to atmospheric seeing, detector noise, and the use of a fixed pixel measurement during the drift scan.

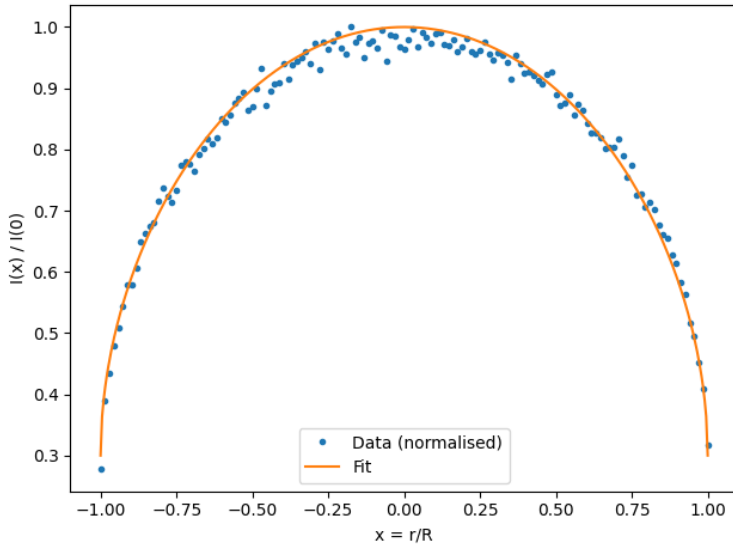


Figure 3: Normalised centre-to-limb intensity profile $I(x)/I(0)$ for the Blue filter, with the best-fit limb darkening model from Eq. 2 overlaid.

The resulting best-fit parameters for the Blue filter are summarised in Table ???. The quoted uncertainties represent one standard deviation errors derived from the covariance matrix.

Filter	λ (nm)	a_0	a_1
UV	350	0.2189 ± 0.0385	0.4012 ± 0.1185
Blue	445	0.3002 ± 0.0090	0.6064 ± 0.0109
Green	538	0.2924 ± 0.0979	0.2936 ± 0.3041
Red	638	0.2264 ± 0.0816	0.7642 ± 0.2514
Infrared	—	0.2999 ± 0.0382	0.9430 ± 0.1177

Table 1: Best-fit limb darkening coefficients for each wavelength band. Uncertainties are one standard deviation errors from the fit covariance matrix.

5 Source Function and Temperature

The fitted limb darkening profiles can be used to infer the source function as a function of optical depth. Under the Eddington–Barbier approximation, the emergent intensity at viewing angle $\mu = \cos \theta$ is approximately equal to the source function evaluated at optical depth $\tau = \mu$,

$$I(\mu) \simeq S(\tau = \mu), \quad (3)$$

where the viewing angle is related to the disc coordinate by

$$\mu = \sqrt{1 - x^2}, \quad x = \frac{r}{R}. \quad (4)$$

The limb darkening model fitted in Section 4 can be written in terms of μ as

$$\frac{I(\mu)}{I(1)} = a_0 + a_1 \mu + (1 - a_0 - a_1) \mu^2, \quad (5)$$

where $I(1)$ corresponds to the disc centre intensity. Combining Eqs. 3–5 gives an expression for the source function profile as a function of optical depth,

$$\frac{S(\tau)}{S(1)} = a_0 + a_1 \tau + (1 - a_0 - a_1) \tau^2, \quad 0 \leq \tau \leq 1. \quad (6)$$

Assuming local thermal equilibrium (LTE), the source function is equal to the Planck function, and over the wavelength range considered the source function scaling may be used to infer the temperature structure. We therefore take the temperature ratio to follow the source function ratio,

$$\frac{T(\tau)}{T(1)} \simeq \frac{S(\tau)}{S(1)}. \quad (7)$$

To obtain absolute temperatures, the profile was normalised using a reference disc centre temperature $T(1) = 5778$ K. The resulting temperature profiles for all observed wavelengths are shown in Figure 4. The reconstructed temperature decreases towards smaller optical depth, consistent with radiation emerging from higher, cooler layers when viewed closer to the limb. Differences between wavelengths reflect the wavelength dependence of the opacity and hence the depths sampled in the photosphere.

6 Discussion

The centre to limb intensity profiles obtained in this experiment exhibit clear and systematic limb darkening at all observed wavelengths. The fitted limb darkening coefficients show a strong dependence on wavelength, consistent with theoretical expectations for radiative transfer in the solar photosphere. In particular, the degree of limb darkening is generally stronger at shorter wavelengths, showcasing the fact that radiation at shorter wavelengths is formed higher in the atmosphere, where the temperature is lower.

The reconstructed temperature profiles derived from the source function further illustrate this behaviour. For all wavelengths, the temperature decreases monotonically with decreasing optical depth, as expected for a stratified stellar atmosphere in radiative equilibrium. The absolute temperature scale, normalised to a disc centre temperature of 5778 K, produces temperature gradients that are broadly consistent with standard solar atmosphere models. Differences between the temperature profiles at

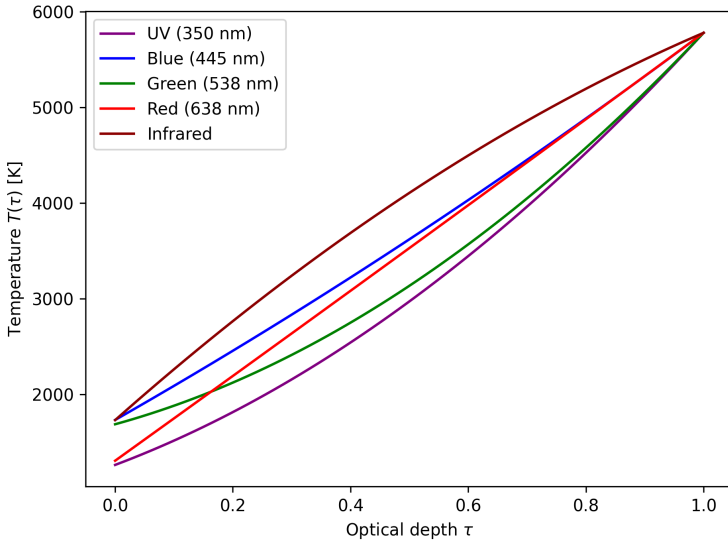


Figure 4: Reconstructed photospheric temperature profiles as a function of optical depth for all observed wavelengths. The curves were computed using Eq. 6 with fitted coefficients (a_0, a_1) for each filter and converted to absolute temperature via Eq. 7 using $T(1) = 5778$ K.

different wavelengths reflect the wavelength dependence of the opacity, with shorter wavelengths probing higher, cooler layers of the photosphere and longer wavelengths sampling deeper, hotter regions.

Uncertainties in the fitted limb darkening coefficients vary between filters, with larger uncertainties generally observed for the Green and Red bands. This is likely due to a combination of reduced limb contrast at longer wavelengths, atmospheric seeing effects, and the use of a single fixed pixel during the drift scan. Small asymmetries between the ingress and egress portions of the drift scans may also contribute to scatter in the measured intensity profiles. Despite these effects, the overall trends remain robust and are well reproduced by the fitted models.

Several limitations of the experimental method should be noted. The use of a single pixel measurement, while effective in avoiding vignetting, increases sensitivity to seeing-induced image motion and detector noise. Averaging over a small region of pixels would reduce statistical noise and provide more reliable uncertainty estimates. Additionally, the frame to coordinate mapping assumes a constant drift rate across the detector, which may be affected slightly by tracking imperfections or atmospheric refraction.

More sophisticated modelling approaches could further improve the analysis. In particular, inclusion of a constant background term and convolution with the telescope point spread function would allow a more accurate treatment of instrumental effects near the limb. Repeating the observations under different seeing conditions or performing multiple drift scans per filter would also improve the statistical robustness of the results.

Overall, the results demonstrate that simple drift scan observations combined with parametrised limb darkening models provide a powerful and accessible method for probing the thermal structure of the solar photosphere.

7 Extensions and Further Analysis

While the parametrised limb darkening analysis presented in this report captures the dominant features of the centre to limb intensity variation, several extensions may be considered to further improve the physical realism and validity of the results.

7.1 Instrumental Effects and Point Spread Function

The limb darkening model used in this analysis assumes that the observed intensity profile directly reflects the intrinsic solar intensity distribution. In practice, the measured profile is modified by instrumental effects, most notably convolution with the telescope point spread function (PSF). Near the solar limb, PSF smearing can artificially smooth sharp intensity gradients, leading to a systematic underestimation of the true limb darkening strength.

A more complete treatment would involve convolving the theoretical limb darkening profile with an appropriate PSF model prior to fitting. The PSF may be approximated by a Gaussian profile characterised by a width related to the atmospheric seeing and telescope aperture. Incorporating PSF convolution into the fitting procedure would allow instrumental broadening effects to be separated from the intrinsic solar limb darkening, improving the accuracy of the fitted coefficients, particularly near the limb where deviations are most pronounced.

7.2 Alternative Uncertainty Estimation and Fitting Approaches

In this work, uncertainties on the fitted limb darkening coefficients were estimated using the covariance matrix returned by the least-squares fitting routine. While this approach captures statistical uncertainties arising from detector noise and fitting residuals, it does not fully account for systematic effects such as seeing-induced image motion or uncertainties in the frame to coordinate mapping.

An alternative approach would be to perform a bootstrap or Monte Carlo resampling analysis, in which synthetic intensity profiles are generated by distorting the measured data within their estimated uncertainties. Repeating the fitting procedure across many such realisations would yield a more reliable estimate of parameter uncertainties and correlations. Additionally, fitting the ingress and egress portions of the drift scan independently would provide a useful cross check for asymmetries and systematic biases.

7.3 Comparison with Standard Solar Atmosphere Models

The reconstructed temperature profiles may also be compared with established semi empirical solar atmosphere models, such as the VAL or HSRA models. While a detailed quantitative comparison lies beyond the scope of this project, the overall temperature gradients obtained here are qualitatively consistent with these models, which predict a monotonic decrease in temperature with decreasing optical depth throughout the photosphere.

Differences between the reconstructed profiles and model predictions may arise from the simplifying assumptions adopted in this analysis, particularly the use of the Eddington–Barbier approximation and the assumption of local thermal equilibrium. Deviations from LTE and wavelength dependent opacity effects become increasingly important in the upper photosphere, especially

at shorter wavelengths. Incorporating more sophisticated radiative transfer modelling would allow these effects to be explored in greater detail.

Overall, these extensions highlight potential avenues for improving the analysis and demonstrate that even simple observational techniques can be connected meaningfully to more advanced theoretical models of stellar atmospheres.

8 Conclusion

In this project, centre to limb intensity variations across the solar disc were measured using drift scan observations at multiple wavelengths. Parametrised limb darkening models were successfully fitted to the measured intensity profiles, allowing the wavelength dependence of solar limb darkening to be quantified through fitted coefficients.

Using the Eddington–Barbier approximation and assuming local thermal equilibrium, the fitted limb darkening profiles were used to reconstruct the source function and derive absolute temperature profiles as a function of optical depth. The resulting temperature gradients exhibit the expected decrease towards smaller optical depths, consistent with radiation emerging from higher, cooler layers of the solar photosphere.

Comparison between wavelengths demonstrates that shorter wavelengths probe higher atmospheric layers, while longer wavelengths sample deeper, hotter regions, in agreement with standard models of radiative transfer in stellar atmospheres. Despite limitations arising from atmospheric seeing, detector noise, and the use of a single pixel drift scan method, the overall trends are ironclad and physically meaningful.

This investigation demonstrates that relatively simple observational techniques, when combined with appropriate modelling, provide valuable insight into the thermal structure of the solar photosphere.